Experiment -4

Aim: SETTING UP A LOCAL AREA NETWORK (LAN)

IP Address Configuration Table

Device	IP Address	Subnet mask	Connected To
PC0	10.1.1.1	255.0.0.0	Switch Port Fa0/1
PC1	10.1.1.2	255.0.0.0	Switch port Fa0/2
PC2	10.1.1.3	255.0.0.0	Switch port Fa0/3
PC3	10.1.1.4	255.0.0.0	Switch port Fa/4
Switch	10.1.1.5	255.0.0.0	VLAN1 Interface

Implementation in Cisco Packet Tracer

Step1: Place the Devices

1. Open cisco Packet tracer
2. From the device category Panel, select End Device and drag 4 onto the workspace
3. Select switches and drag 1 switch

Step2: Cable the Devices

4. Click connections → select copper straight through cable
5. Click each PC then click switch, and select an available Fast Ethernet port to complete each connection
6. Repeat for all four PCs.

Step 3: Assign IP Address to PC

7. Click a PC → Desktop tab → IP configuration
8. Enter the static IP Address and subnet mask

Step 4: Configure the switch management IP.

9. Click the switch → CLI tab

10. Enter the following commands to assign a management IP:

enable

configure terminal

interface vlan 1

ip address 10.1.1.5 255.0.0.0

no shutdown

exit

Step 5: Verify connectivity using Pings

11. Click a PC → Desktop tab → command Prompt

12. Use the 'ping' command to test connectivity to other PCs

13. Successful "Reply from..." message confirms connectivity

Step 6: Demonstrate Data sharing (Server simulation)

14. Add a server device (End Devices → servers) and connect it to the switch

15. Assign it an IP address via its Desktop → IP config tab

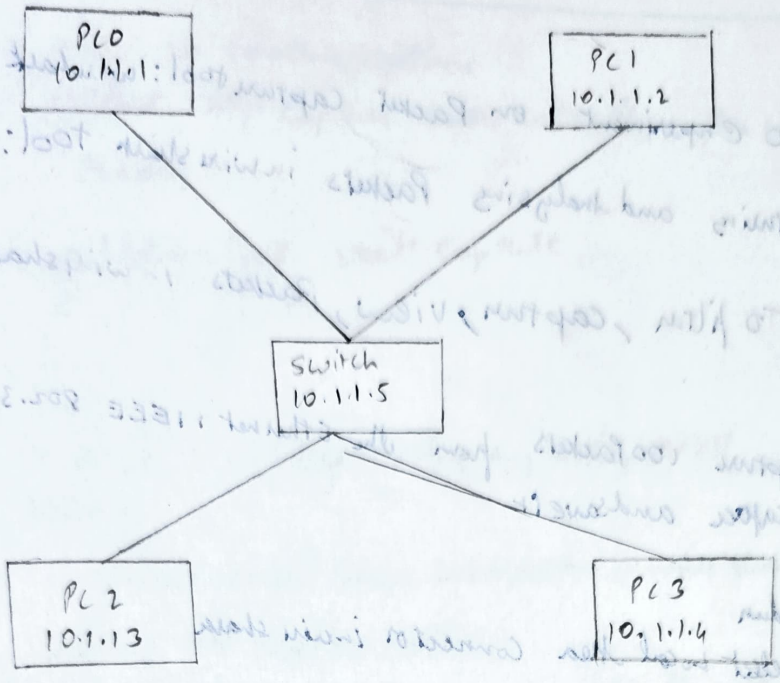
16. Open the server's Services (config) Tab and enable the HTTP service

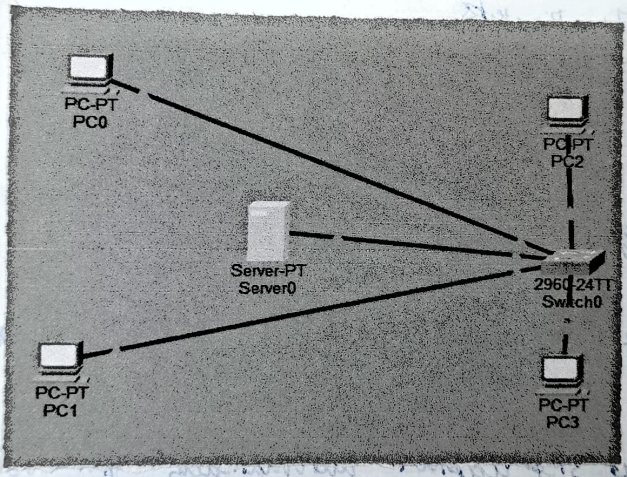
17. From another PC, open the web Browser (Desktop tab) and enter the server's IP address to load

Step 7: Observe

18. Switch to simulation mode

19. Send a Ping or web request between two devices and use the Play/capture control to step through the request





Result: Thus the setting up of Local area network (LAN) has been executed successfully